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SURVEY

Prerequisite Relation Learning: A Survey and Outlook

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Prerequisite Relation Learning: A Survey and Outlook

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Prerequisite relation (PR) learning is a fundamental task in educational technology that identifies dependencies between learning resources to facilitate personalized learning experiences and optimize educational content organization. This survey provides a systematic review of prerequisite relation learning, emphasizing both methodological advances and practical applications. We first explore two distinct granularities of learning resources: knowledge concepts (KCs) and learning objects (LOs), establishing their definitions and relationships. We then introduce a novel classification framework for prerequisite relation learning methods based on both feature types and enhancement relationships, categorizing existing approaches into four types: (1) multi-source knowledge features for KCs' prerequisite relation learning; (2) semantic knowledge features for LOs' prerequisite relation learning; (3) LOs-enhanced learning for KCs' prerequisite relation learning; and (4) KCs-enhanced learning for LOs' prerequisite relation learning. The survey highlights recent developments in modeling KCs' prerequisite relations. We provide a comprehensive analysis of evaluation methodologies, including both intrinsic metrics and extrinsic evaluation. Furthermore, we analyze the practical impact of prerequisite relations in educational applications, from adaptive learning path generation to curriculum design. Finally, we discuss current challenges and future opportunities for prerequisite relation learning.

CCS Concepts: • **Applied computing** → **Education**; • **Computing methodologies** → **Knowledge representation and reasoning**;

Additional Key Words and Phrases: Prerequisite relation, knowledge concept, learning object, concept graph learning, Wikipedia

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1 Introduction

Prerequisite relation (PR) represents the inherent dependencies among learning resources.¹ These dependencies play a crucial role in key cognitive processes: learning, organizing, applying, and generating knowledge [50]. Prerequisite relations ensure that learners approach new topics equipped with the necessary prior knowledge, thereby facilitating comprehension and boosting confidence in the subject matter [52, 55]. Considering two KCs, which represent atomic units of knowledge within educational contexts, A and B , we define $A \rightarrow B$ to signify that A is a prerequisite of B . While PR can be formally defined, its implementation in educational practice is more complex. For instance, in an elementary algebra course, while the KC of *variables* is a prerequisite to the KC of *equations*, educators may approach this either by introducing abstract variable concepts first or by starting with concrete equation examples. This flexibility allows teachers to adapt prerequisite relations to their specific instructional needs.

Identifying prerequisite relations is fundamental to educational content organization. These relations help establish logical knowledge structures in which foundational knowledge precedes advanced topics. PR has demonstrated substantial value in several key educational applications. For example, learning path personalization often leverages PR as one approach to organize content sequences [59, 90, 91]. PR-based course material organization enhances curriculum planning by ensuring logical knowledge progression [19, 57, 72, 95]. For reading materials, PR facilitates the generation of effective reading lists by identifying and organizing content dependencies [37, 47, 92]. In learning analytics, students' performance patterns can help infer prerequisite structures [15, 23, 61, 64], which in turn support student ability profiling and cognitive diagnosis [33] by identifying fundamental knowledge gaps affecting advanced knowledge understanding [43, 65].

The learning of prerequisite relations in educational settings focuses on predicting inherent dependencies between learning resources, determining the necessary sequence of mastery. This challenge can be effectively formulated as a link prediction problem, leveraging methodologies and models from complex knowledge networks. Link prediction has long been a key area of research in network science, attracting interest from various disciplines due to its wide-ranging applicability. Its foundations stem from early data mining studies in computer science, as demonstrated by Sarukkai et al.'s pioneering work using Markov chains to predict links and analyze paths on the web [87]. Subsequently, Liben-Nowell and Kleinberg examined its applications in social cooperation networks [58]. Today, link prediction techniques are widely applied in various domains such as social networks [27, 41, 101], bioinformatics networks [16, 49, 62], and citation networks [39, 89, 118]. Building on these advances, our work specifically explores link prediction within knowledge networks [22, 73], particularly in the context of predicting prerequisite relations between learning resources.

The exponential growth of online educational platforms has dramatically increased the availability of diverse internet-based learning resources, transforming the landscape of digital education. However, this rapid growth has presented a significant challenge: establishing effective PR among these educational materials [84, 96]. Educational resources originate from multiple sources, including structured MOOC curricula, course materials, and user-contributed content like Wikipedia and tutorials. Furthermore, learners show diverse learning behaviors, including varying acquisition speeds, different background knowledge, and individual preferences for learning sequences, which complicates identifying prerequisite relations for all learners. A key difficulty arises from how these varied resources convey similar concepts: different levels and perspectives reveal that prerequisite relations are not universal [55]. The same KCs may follow

¹Learning resources include various formats such as knowledge concepts, educational documents, and web URLs, among others.

different prerequisite structures depending on the educational material and context. For instance, foundational KCs may receive comprehensive coverage in certain materials while appearing as prerequisite knowledge in others, potentially creating knowledge gaps for learners.

The study of prerequisite relations within educational resources has emerged as a crucial response to these pedagogical challenges. This task aims at developing automated methods for identifying dependency relationships among learning materials, thereby establishing rational knowledge progression paths and coherent educational trajectories for educational platforms. Recent research has increasingly used machine learning and deep learning techniques in analyzing knowledge networks and optimizing the structural organization of online learning platforms [53, 55, 96]. The task continues to evolve toward more sophisticated models capable of real-time adaptation and dynamic prerequisite mapping, in order to better accommodate the ever-changing nature of educational content and the diverse needs of learners.

While prerequisite relation learning has attracted substantial research attention, a comprehensive technical review of this task remains absent from the literature. This survey presents a systematic review of prerequisite relation learning methods and their technical evolution since 2012. We introduce key methodologies pertaining to prerequisite relations through a structured evaluation of seminal contributions in the PR task. Additionally, we summarize the datasets and evaluation metrics commonly used in previous studies and conclude by discussing potential future challenges and opportunities for research. Our survey provides both broad coverage (examining prerequisite relations for KCs and LOs) and deep insights (through detailed classification and comparative analysis). This survey aims at serving as an authoritative technical reference for researchers and practitioners in this task. As shown in Figure 1, our main contributions through this survey are outlined as follows:

- This survey provides a systematic review of existing approaches for identifying prerequisite relations, highlighting key advancements and identifying technical limitations in the current literature.
- Through a comprehensive analysis of representative methods, we examine their methodological foundations and practical applications. We further analyze their underlying assumptions, technical designs, and empirical effectiveness to offer insights for future research directions.
- This survey presents a detailed summary of commonly used datasets and evaluation metrics in the PR task. We also discuss both unsolved challenges and future opportunities, with particular focus on methodological limitations and potential avenues for improvement.

This survey is structured to provide a comprehensive overview of prerequisite relation learning. Section 2 establishes the theoretical foundation by clarifying the relationship between KCs and LOs, examining the core definitions and properties of prerequisite relations, and presenting various classification criteria. Section 3 systematically categorizes existing prerequisite relation learning methods into four distinct approaches. Section 4 offers a detailed analysis of publicly available datasets and evaluation metrics in this task. Section 5 discusses ongoing challenges and future opportunities in prerequisite relation learning. Finally, Section 6 concludes with the main insights into prerequisite relation learning and its potential applications.

2 Preliminaries

2.1 Research Objects: Knowledge Concepts and Learning Objects

In prerequisite relation learning, two fundamental types of learning resources are distinguished based on their levels of granularity [12, 28]: **Knowledge Concepts (KCs)** and **Learning Objects (LOs)**. KCs can be formally represented as a set $K = \{k_1, k_2, \dots, k_n\}$, where each k_i denotes an atomic unit of knowledge within its specific educational context [48]. For example, in

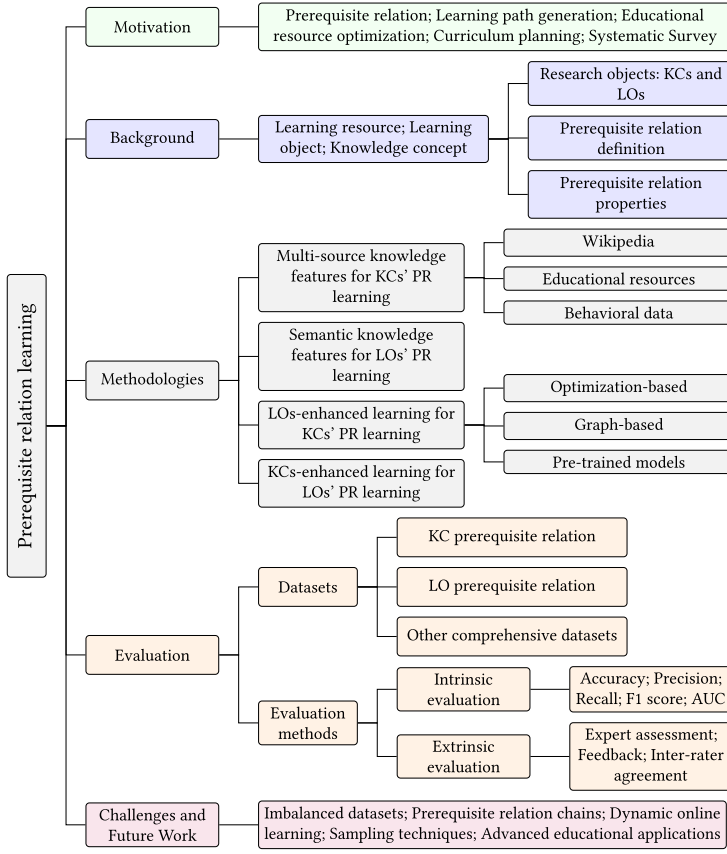


Fig. 1. Overview of the survey.

mathematics, KCs could include *fraction*, *improper fraction* and *decimal fraction*, while in programming, KCs could include *variable declaration*, *loop structures*, and *function calls*. These KCs form the basic building elements for LOs, which operate at a coarser granularity and can be denoted as $O = \{o_1, o_2, \dots, o_m\}$, where each o_i represents a self-contained educational unit that typically encapsulates multiple KCs. For instance, a LO titled “Converting between Fractions and Decimal” might contain KCs related to *proper fractions*, *decimal notation* and *percentage understanding*. The IEEE Learning Technology Standards Committee formally defines a LO as ‘any entity, digital or non-digital, that may be used for learning, education, or training’ [83].

The relationship between KCs and LOs can be formalized through a mapping function $M : O \rightarrow P(K)$, where $P(\cdot)$ denotes the power set of KCs, indicating that each LO maps to a subset of KCs it encompasses. As illustrated in Figure 2, prerequisite relations primarily manifest at two levels in this framework: (1) KC-to-KC prerequisites, represented as directed edges between KCs within an LO, which demonstrate fundamental concept dependencies (e.g., understanding *fractions* is prerequisite to learning *decimal fractions*); and (2) LO-to-LO prerequisites, shown as connections between learning objects, indicate module-level dependencies (e.g., O_1 must precede O_2). These two types of prerequisite relations underpin different learning approaches. KC-level prerequisite learning emphasizes fine-grained concept relationships, often inferred from domain knowledge graphs or educational data. In contrast, LO-level prerequisite learning typically leverages semantic features, content structure, and metadata within educational resources.

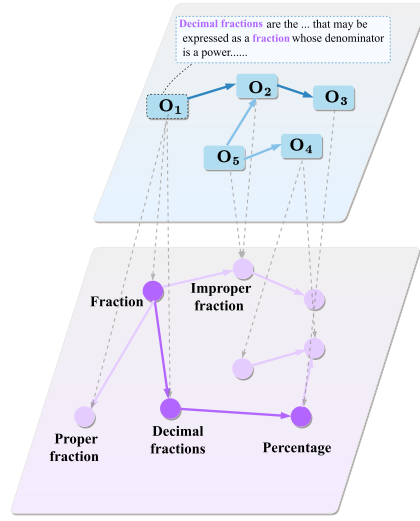


Fig. 2. An illustrative example of prerequisite relations in mathematics education showing dual-layer dependencies: the LOs-level (blue rectangles) containing “Decimal fraction” material content with prerequisite relations in the upper layer, and their mapped KCs (purple circles) with KC-level prerequisite relations in the lower layer.

This multi-level representation of educational content plays a crucial role in modern educational systems, particularly in adaptive learning environments and curriculum design. At the KC level, accurate prerequisite relations enable precise optimization of learning paths, ensuring students master fundamental KCs before advancing to more complex ones [95]. At the LO level, prerequisite relations guide the organization of larger educational modules, such as determining the optimal sequence of reading lists and MOOC lecture materials [57, 76]. However, identifying these prerequisite relations poses significant challenges due to the diverse formats of educational resources and the complex relationships between KCs and LOs. These challenges have fueled extensive research into prerequisite relation learning, leading to the development of methods that range from multi-source feature extraction and semantic representation to enhanced learning approaches that exploit the structural interplay between KCs and LOs.

2.2 Prerequisite Relation

2.2.1 Prerequisite Relation Definition. Building upon the previously defined notions of KCs and LOs, we define prerequisite relations between two learning resources (KCs and LOs) as follows: given a pair of learning resources (A, B) , if understanding or mastering A is necessary for effectively learning or understanding B , then A is considered a prerequisite of B , denoted as $A \rightarrow B$ [56, 96]. This relationship can be formalized through a function $\text{Preq}(A, B)$:

$$\text{Preq}(A, B) = \begin{cases} 1, & \text{if } A \text{ is a prerequisite of } B \\ -1, & \text{if } B \text{ is a prerequisite of } A \\ 0, & \text{else} \end{cases} \quad (1)$$

The task of prerequisite relation learning is predominantly framed as a binary classification problem, where the objective is to predict whether A is a prerequisite for B (i.e., $\text{Preq}(A, B) = 1$).

Some research [96, 125] has extended the task into a three-class classification task, distinguishing between positive prerequisites ($\text{Preq}(A, B) = 1$), negative prerequisites ($\text{Preq}(A, B) = -1$), and

non-prerequisites ($\text{Preq}(A, B) = 0$). This extension moves beyond the conventional binary classification paradigm, enabling the modeling of more nuanced pedagogical relationships and dependencies. In addition, some approaches have proposed continuous representations of prerequisite relations. Notably, methods such as RefD [55] output continuous scores between 0 and 1, reflecting the strength or confidence of prerequisite relations. These scores can then be converted to binary decisions using threshold tuning.

Although prerequisite relations play a crucial role in organizing learning sequences, the precise connections between KCs or LOs depend significantly on pedagogical approaches [17]. Different educational strategies implement KCs/LOs in distinct sequential arrangements based on their pedagogical methodologies, learning contexts, and learner characteristics [36]. This contextual variability indicates that prerequisite relations are not universally fixed across different educational settings, making their evaluation and validation particularly challenging. In this survey, we focus on methods that assume static prerequisite relations, aiming at uncovering generalizable patterns through computational approaches.

2.2.2 Prerequisite Relation Properties. The prerequisite relation between learning resources establishes a fundamental pedagogical dependency, in which the mastery of prerequisite knowledge serves as essential foundational knowledge for subsequent learning objectives. These relations form a hierarchical structure governed by three core principles that facilitate systematic knowledge acquisition and cognitive progression [4, 5, 111].

- (1) **Non-reflexivity:** A learning resource cannot be a prerequisite for itself, ensuring that prerequisite relations occur only between distinct learning entities. This prevents self-loops and promotes a logically progressive educational experience [4, 5, 56].
- (2) **Anti-symmetry:** For any pair of learning resources, (A, B) , if A is a prerequisite for B , then B cannot be a prerequisite for A . This property prevents circular dependencies in the learning sequence, ensuring a unidirectional educational trajectory [5, 111].
- (3) **Transitivity:** Given three distinct learning resources A, B , and C , if A is a prerequisite for B and B is a prerequisite for C , then A is also a prerequisite for C . This principle reinforces the hierarchical organization of knowledge, promoting mastery of foundational concepts before advancing to more complex material [96, 110].

These properties are crucial for constructing prerequisite relation datasets. In most domains, prerequisite relations are inherently sparse, with only a small fraction of possible resource pairs exhibiting actual dependencies. Enforcing these properties during dataset construction not only enhances structural coherence but also improves the robustness and generalizability of models trained on such data, leading to more accurate and reliable predictions.

2.3 Classification Basis

The classification of prerequisite relation learning methods is based on the nature of features and the enhancement relationship between different types of learning resources. This classification aims at systematically organizing existing approaches and clarify their methodological focuses and application scenarios. This section builds upon the foundational concepts introduced in Sections 2.1 and 2.2, where we discussed the types of learning resources (KCs and LOs) and the relationships between them. Here, we categorize methods according to how these resources are utilized and combined to learn prerequisite relations.

2.3.1 Nature of Features. The classification of prerequisite relation learning methods can be understood by examining the nature of the features extracted for prerequisite learning. There are primarily two types of features used:

- (1) **Multi-source knowledge features for KCs:** These features are typically derived from various data sources, including Wikipedia, textbooks, and online educational platforms (e.g., structured course materials and learner activity logs). Methods in this category focus on integrating structural and semantic features from multiple sources to enhance the understanding of prerequisite relations at a fine-grained level. The subcategories under this classification are based on different approaches in combining and analyzing these diverse data sources:
 - **Link-based features:** These methods exploit hyperlink structure and structural features of sources like Wikipedia to capture direct prerequisite relations between KCs.
 - **Semantic features:** Derived from educational materials like textbooks or online learning materials, these features focus on understanding the deeper meanings and relations between KCs using **natural language processing (NLP)** techniques.
 - **Behavioral features:** These approaches incorporate data from user interactions, such as click-stream, content navigation patterns, and question-response records, to infer prerequisite structures based on how learners interact with educational content.
- (2) **Semantic knowledge features for LOs:** LOs represent broader learning units that encompass multiple KCs. Semantic features extracted from LOs, such as lecture notes, instructional videos, and course materials, provide rich contextual representation that aids in establishing prerequisite relations at a modular level. These methods focus on capturing the meaning and context of LOs to model their prerequisite relations. The subcategories in this context are based on the specific type of LOs and the methodologies used to extract semantic information:
 - **Text-based learning objects:** These include lecture notes, textbooks, and slides, where NLP techniques are applied to extract topic structures and content relationships.
 - **Multimedia-based learning objects:** For LOs such as videos, methods often use multi-modal data, including audio, video transcripts, and user interaction logs, to understand content and establish relationships.

2.3.2 Enhancement Relationships. Another dimension of classification is based on whether and how the learning of one type of resource is enhanced by another:

- (1) **LOs-enhanced learning for KCs:** In this category, LOs are used to enhance the learning of KCs by providing auxiliary information. This integrated approach allows for a deeper understanding of the prerequisite relations between KCs by utilizing the rich information available in associated LOs. The methods in this category can be further divided into three subcategories based on the techniques used to enhance KCs' PR learning:
 - **Optimization-based methods:** These methods leverage optimization techniques to infer KC relationships using course-level dependencies as proxies. The classification is based on the use of optimization to align LOs with KC-level relations effectively, ensuring logical consistency across dependencies.
 - **Graph-based models:** These approaches use heterogeneous graphs that incorporate both LOs and KCs to learn prerequisite relations. The use of graph models allows for capturing local and global dependency patterns, providing a more integrated view of the relationships. The subcategory is motivated by the ability of graph-based techniques to represent complex networks of LOs and KCs.
 - **Pre-trained models:** These methods utilize pre-trained language models, such as BERT [26], to capture semantic relationships from LOs while also incorporating structural information from knowledge graphs. The classification arises from the dual use of both

Table 1. Multi-source Knowledge Features for KCs' PR Learning

Data Source	Feature Types	Related Works
Wikipedia	Link structure	[7, 67, 79, 96, 109, 110]
	Reference patterns	[6, 55, 75, 125], [3, 42, 56]
	Semantic content	[10, 11, 56, 66, 85, 103, 110]
Education resources	Textbook	[2, 74, 77, 103]
	Courses/MOOCs	[9, 75, 99, 109]
	Lectures/Slides	[34, 37, 53]
Behavioral data	Click-stream logs	[24, 40, 88, 108]
	Learning trajectories	[98, 117]
	Assessment responses	[20, 23, 32, 44, 78, 120]

pre-trained contextual representations and structural graph information to enhance performance accuracy.

- (2) **KCs-enhanced learning for LOs:** Conversely, KCs are used to enhance the learning of LOs by providing a structured understanding of their components. The prerequisite relations between LOs can be inferred from the established dependencies of their constituent KCs. This approach facilitates a more granular analysis of LOs and helps in generating structured learning paths. The subcategories are defined by how KCs are used to influence LO-level relationships:
 - **Dependency-based inference:** This subcategory uses established KC dependencies to directly infer LO-level prerequisite relations, ensuring a logical progression through learning modules.
 - **Cross-domain transfer:** This subcategory focuses on using KC relationships to facilitate prerequisite learning across different domains or educational contexts, highlighting the generalizability of KC-based dependencies.

These classification criteria not only help in organizing existing research but also in identifying gaps and opportunities for further advancements in the task. By distinguishing approaches based on feature types and enhancement relationships, researchers can better understand the unique challenges and opportunities associated with each method and apply them effectively in different educational contexts.

3 Prerequisite Relation Learning Methods

In this section, we categorize existing research on prerequisite relation learning into four categories: (1) Multi-source knowledge features for KCs' PR learning (Section 3.1); (2) Semantic knowledge features for LOs' PR learning (Section 3.2); (3) LOs-enhanced learning for KCs' PR learning (Section 3.3); (4) KCs-enhanced learning for LOs' PR learning (Section 3.4). Tables 1 and 2 summarize the main works that implement these various methods for KCs' prerequisite relation learning.

3.1 Multi-source Knowledge Features for KCs' PR Learning

The inherent complexity of KCs necessitates a multi-source approach that integrates evidence from knowledge bases, educational resources, and behavioral data to fully capture prerequisite relations. As illustrated in Figure 3 and Table 1, this multi-source approach combines features from Wikipedia (providing structural and semantic knowledge), educational resources (including textbooks, MOOCs, and slides), and behavioral data (such as click data, learning logs, and test records) to comprehensively model prerequisite relations.

Table 2. LOs-enhanced Learning for KCs' PR Learning

Method	Key Characteristics	Related Works
Optimization-based	Course dependency integration Structure-based inference	[57, 63]
Graph-based	Heterogeneous graph modeling Multi-relation representation	[42, 52, 54, 121, 121, 122]
Pre-trained models	Deep semantic understanding Contextual representation learning	[29, 51, 93, 94, 97, 104, 126]

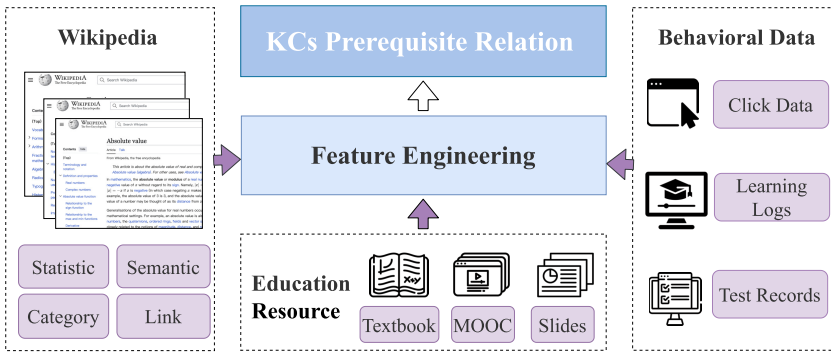


Fig. 3. Multi-source data integration for feature engineering in KCs' prerequisite relation learning. The framework illustrates how Wikipedia, education resources, and behavioral data are integrated to enable feature engineering for prerequisite relation learning.

3.1.1 Features from Wikipedia. Wikipedia establishes comprehensive knowledge networks through its hyperlink system and knowledge organization. In this knowledge base, KCs are connected with their related KCs via hyperlinks. For example, the KC of *multiplication*² is often linked with related KCs like *addition* and *subtraction*. These connections form a directed graph where nodes represent KCs and edges indicate their cite relationships. Such knowledge concept network characteristics offer valuable insights for prerequisite relation learning.

Building upon this link structure, researchers have developed machine learning methods to capture implicit prerequisite relations between KC pairs. Talukdar and Cohen [96] were the first to formalize prerequisite relations within Wikipedia as a directed graph structure. In their work, they analyzed KC pairs using multiple Wikipedia features (hyperlinks, edit histories, and page content) and employed a **maximum entropy (MaxEnt)** classifier to identify these relations. Inspired by this feature-based approach, subsequent research has extended the analysis by introducing advanced features extracted from Wikipedia knowledge bases. These features fall into three categories: (1) structural features (e.g., node degrees, KCs category and neighbor counts); (2) semantic relatedness metrics; (3) structural similarity indicators, each contributing to more accurate identification of prerequisite relations.

Among various link-based features, reference patterns between KCs in Wikipedia provide particularly valuable signals for prerequisite relations. If most concepts related to concept *A* refer to concept *B*, while few concepts related to *B* refer to *A*, this pattern suggests that *B* is likely to be a prerequisite of *A*. For example, in Wikipedia, concepts related to *calculus* (such as *derivatives*,

²<https://en.wikipedia.org/wiki/Multiplication>

limits, and *rates of change*) often refer to *algebra*, while concepts related to *algebra* (such as *equations*, *variables*, and *functions*) rarely require references to *calculus*, indicating that *algebra* is a prerequisite for *calculus*. Based on this observation, Liang et al. [55] introduced a link-based metric called **reference distance (RefD)**. RefD quantitatively models prerequisite relations by evaluating the asymmetry of reference patterns between Wikipedia KCs:

$$\text{RefD}(A, B) = \frac{R_{B,A}}{I_B} - \frac{R_{A,B}}{I_A}, \quad (2)$$

where I_B represents the number of related concepts to KC B , and $R_{B,A}$ represents the number of related concepts of B that cite KC A . Therefore, $\frac{R_{B,A}}{I_B}$ indicates the proportion of related concepts of B that cite KC A . Similarly, $\frac{R_{A,B}}{I_A}$ represents the proportion of related concepts of A that cite KC B . The authors suggest that when the value of $\text{RefD}(A, B)$ is close to 1, it indicates that the related concepts of B cite A much more frequently than the related concepts of A cite B , implying that A is likely a prerequisite for B . Conversely, when the value of $\text{RefD}(A, B)$ is close to -1, B is likely a prerequisite for A . To classify these continuous values, a threshold mechanism is introduced:

$$\text{RefD}(A, B) \in \begin{cases} (\theta, 1], & \text{if } A \text{ is a prerequisite of } B \\ [-\theta, \theta], & \text{if no prerequisite relation} \\ [-1, -\theta), & \text{if } B \text{ is a prerequisite of } A \end{cases}, \quad (3)$$

where θ is a positive threshold constrained to $[0, 1)$. The optimal value of θ is calculated through empirical experiments to maximize the accuracy of prerequisite relation identification.

RefD, as an advanced feature engineering implementation, has demonstrated several advantages over previous methods: (1) incorporating neighboring KCs information for rich contextual understanding, which extends beyond traditional link frequency analysis used in early approaches; (2) extending KC link reference coverage through related KCs when direct links are sparse, addressing a key limitation of traditional link-based features. These characteristics, along with its quantifiable metrics for systematic threshold optimization, have established RefD as a robust feature for learning of prerequisite relations.

However, despite the effectiveness of link-based features in capturing structural relationships, they cannot model semantic relationships hidden in Wikipedia articles. To address this limitation, researchers have explored semantic features derived from Wikipedia article content through DL architectures. Miaschi et al. [70] proposed a hybrid approach that used LSTM networks fed with both manually crafted features and deep representations extracted from KC pairs. The evolution continued through the EVALITA 2020 competition, where state-of-the-art models such as XGBoost, GRU, and BERT were employed to process both semantic features and link-based features. The combination of structural and semantic features represents a comprehensive approach to modeling prerequisite relations between KCs, leveraging both Wikipedia's link structural knowledge and article content.

3.1.2 Features from Education Resources. While Wikipedia features focus on general knowledge relationships, educational resources offer unique advantages through their systematic design by domain experts. These resources provide structured instructional sequences and comprehensive assessment data for validating learning effectiveness. This empirical and pedagogical foundation contributes significantly to prerequisite relation learning.

Researchers have explored various features from textbooks and course syllabi to identify prerequisite relations. Two main approaches have emerged: joint optimization methods and burst analysis methods. Wang et al. [103] proposed a joint optimization approach that combines KC extraction with prerequisite identification, leveraging both Wikipedia knowledge and textbook

structural information. The proposed framework optimizes these two tasks jointly rather than separately, utilizing features like catalog organization and **table of contents (TOCs)** distance between KCs to achieve better performance. While this approach offers rich features and robust performance, it depends on external knowledge bases. In contrast, Adorni et al. [2] developed an unsupervised method based on burst analysis, which identifies dense intervals of KCs in text and applies interval algebra to infer prerequisite relations. This burst analysis approach analyzes temporal patterns between concept bursting periods using Allen's interval algebra, making it simpler to implement and independent of external knowledge, though more sensitive to textbook writing styles.

The transformation of educational resources from traditional textbooks to online platforms has enabled researchers to exploit various MOOC course materials, including video subtitles, lecture notes [53], and digital slides [30], to explore KCs' prerequisite relations. Pan et al. [75] proposed a machine learning model by combining MOOC video transcripts with Wikipedia knowledge, analyzing KCs' prerequisite relations through semantic relatedness, contextual reference patterns and KCs' structural characteristics from subtitles. Similarly, Xiao et al. [109] developed a hybrid framework integrating both course-based features (e.g., KC frequency, ordering and learning sequences) and Wikipedia-based features (e.g., category hierarchies and user log data) for KCs' prerequisite learning. Both methods demonstrate strong performance through comprehensive feature engineering and knowledge base integration, though this effectiveness is contingent upon the availability of Wikipedia resources.

In contrast to these knowledge-enhanced approaches, some researchers focused on developing methods that exclusively leverage course content itself without label annotation. Aytekin et al. [9] proposed an unsupervised approach using course video subtitles, where prerequisite relations are measured through phrase similarity scores and video platform ranking models. More recently, Torre et al. [99] presented another unsupervised method for lecture videos based on KCs' occurrence frequency analysis, incorporating multimodal features including visual cues and lecture transcripts from MOOC video streams. These methods offer greater deployment flexibility and reduced computational overhead, as they operate independently of external resources, though their effectiveness correlates strongly with the quality and structure of course materials. Motivated by the goal of reducing dependency on external knowledge bases, Gordon et al. [37] examined how scientific articles inherently represent prerequisite relations through their citation structures. By analyzing citation patterns and the sequencing of KCs introduced in articles, they demonstrated that, while scientific literature exhibits less pedagogical organization than educational materials, it still offers valuable signals for learning KCs' prerequisite relations.

Despite achieving promising results on specific educational resources, existing approaches remain constrained by single data sources. To address this limitation, Roy et al. [85] proposed the PREREQ model, which systematically synthesizes multiple educational data sources. Specifically, the method leverages a pairwise-link-LDA model [13] to extract latent representations of KCs from both course data and MOOC videos. These representations simultaneously encode semantic relationships between concepts and capture their organizational structure within educational resources. Subsequently, these representations are fed into Siamese neural network [46], which generalizes from known KCs prerequisites to predict new ones. Experimental results demonstrate that PREREQ outperforms existing methods on standard datasets, with the Siamese neural network architecture demonstrating strong generalization capability in prerequisite relation learning [42, 79, 93, 107].

3.1.3 Features from Behavioral Data. Compared to static knowledge bases and educational materials, behavioral data provides a unique perspective for prerequisite relation learning by directly

capturing how KCs are accessed and understood in users' learning contexts. By analyzing temporal order and interaction patterns between KCs, the behavioral data reveals implicit prerequisite structures through three data sources: Wikipedia click-stream patterns reflecting informal knowledge exploration, MOOC interaction traces capturing structured learning processes, and online practice platforms recording problem-solving trajectories.

Wikipedia click-stream data captures users' actual navigation behavior between KCs from Wikipedia articles, reflecting the natural learning paths of knowledge acquisition. Sayyadiharikandeh et al. [88] pioneered the use of such click-stream data to discover KCs' prerequisite relations. They designed a set of features to model click-stream patterns, including direct click count (weight), backward click count (backward weight), and so on. Their approach is based on the hypothesis that if users frequently navigate from concept B to concept A during learning, this may indicate that A is a prerequisite for B . While Hu et al. [40] improved KC pairs' click coverage by incorporating related KC sets, click-stream data from Wikipedia remains relatively simple and may not fully reflect formal learning processes.

Educational platforms, particularly MOOCs, provide more structured and comprehensive behavioral data in formal learning contexts. Yu et al. [117] explored various learner behaviors in MOOC environments and identified four typical video-watching patterns: sequential, cross-course, skipping, and backward watching. These patterns reveal different aspects of knowledge dependencies, with sequential watching indicating a natural learning progression and cross-course watching suggesting a need for knowledge support from other courses. Video-watching patterns provide high-level interaction sequences; the assessment data from learning platforms, such as quiz responses and assignment submissions, enables more precise evaluation of knowledge component mastery, providing a multi-dimensional perspective for prerequisite relation learning.

Unlike previous methods, the learning interaction data directly recorded by **knowledge tracing (KT)** provides valuable evidence for analyzing prerequisite relations between KCs. Researchers have leveraged these student performance records to infer KC prerequisite relations through various approaches. First, Bayesian network approaches allow experts to define KC prerequisites through probabilistic relationships [3, 20]. Additionally, Käser et al. proposed **dynamic Bayesian networks (DBNs)** to model the causal effects between KCs by incorporating arcs between their mastery variables [44]. Performance prediction-based inference methods leverage KT models, particularly **deep knowledge tracing (DKT)** models [78], to analyze KC dependencies through mastery state prediction. Specifically, when a DKT model prediction accuracy on KC B assessment responses improves by incorporating KC A mastery history but not vice versa, this asymmetric predictive relationship suggests that A is a prerequisite of B [23, 120]. Furthermore, statistical approaches construct skill relationship matrices and employ metrics like Cohen's kappa and Phi coefficients [32] to quantify KC dependencies from student performance data.

These KT-based methods can directly identify learning bottlenecks and KCs' prerequisite relations from student assessment data. However, they require large-scale, high-quality performance records and careful handling of data sparsity issues, especially when students' practice patterns are irregular or incomplete.

3.2 Semantic Knowledge Features for LOs' PR Learning

Different from KCs that are typically well-structured knowledge units, the representation learning for LOs plays a crucial role in prerequisite relation learning due to their diverse formats, ranging from textual materials (lecture notes, exercise sets) to multimedia content (video lectures,

interactive simulations). Representation learning methods aim at learning effective features that can capture the semantic relationships between LOs across different modalities and granularities.

To mitigate Wikipedia limitations in adequately representing key KCs within LOs, Gasparetti et al. [34] developed a simple and effective approach for directly predicting prerequisite relations between LOs. By utilizing the FastText model [14], this method generates compact, low-dimensional latent representations of LOs. These representations are subsequently used in a binary classifier to identify prerequisite relations among educational documents. This approach reduces the dependency on explicit concept identification within each LO, enhancing its applicability compared to traditional features-based prediction methods.

For video-based LOs, different approaches have been proposed to handle their multi-modal nature. Chen et al. [24] explored the identification of prerequisite relations among segments of MOOC videos by integrating both behavioral and semantic information. They segmented MOOC videos into 20-second intervals, utilizing user click-stream data and video caption text embeddings, and predicted prerequisite relations using **Recurrent Neural Networks (RNNs)**. Similarly, Thareja et al. [98] developed a feature-based prerequisite relation learning method specifically for academic videos, combining transcript analysis with language modeling to extract pertinent features from video transcripts.

Despite these advances, several challenges remain in LOs' prerequisite relation learning. First, the semantic gap between different types of LOs (e.g., text vs. video) makes it difficult to develop unified representation methods. Second, the granularity variation of LOs, from fine-grained exercise items to complete lecture materials, requires adaptive feature extraction strategies. Third, the lack of large-scale labeled datasets for different types of LOs hinders the development of more sophisticated deep learning approaches.

3.3 LOs-enhanced Learning for KCs' PR Learning

Sections 3.1 and 3.2 have focused on feature-based approaches that extract and analyze characteristics of KCs and LOs independently: either extracting multi-source knowledge features from KCs or capturing semantic features from LOs. However, these feature-based approaches have two significant limitations. First, they analyze KC-KC and LO-LO prerequisite relations in isolation, failing to capture their mutual influences and potential synergies. Second, they typically rely on pairwise feature extraction, making it difficult to discover extended prerequisite paths that are crucial for curriculum planning. To address these limitations, researchers have explored enhanced methods leveraging both KCs and LOs, as shown in Table 2. These approaches are described in the following sections.

3.3.1 Optimization-based Methods with LO Integration. Early attempts to integrate LO information with KC prerequisite learning focused on optimization-based approaches, which aimed at leveraging the rich structural information inherent in course dependencies to infer KC relations. The key insight behind these methods was that course-level prerequisites could serve as a proxy for KC-level prerequisites, since courses that are prerequisites of each other likely contain KCs with similar prerequisite relations.

Liang et al. [57] pioneered this approach by exploring how dependencies between university courses could be utilized to infer KCs' prerequisite relations. The authors collected course descriptions from computer science courses across 11 U.S. universities and manually annotated inter-course dependencies ($\Omega = i \mapsto j_{i,j \in N}$). They then extracted Wikipedia concepts as KCs from these course descriptions using the Wikipedia miner toolkit [71], representing each course as a TF-IDF vector in the KC space. The key innovation was transforming prerequisite relation prediction into an optimization problem that sought to find the optimal KC prerequisite relation matrix $A = (a_{s,t})$

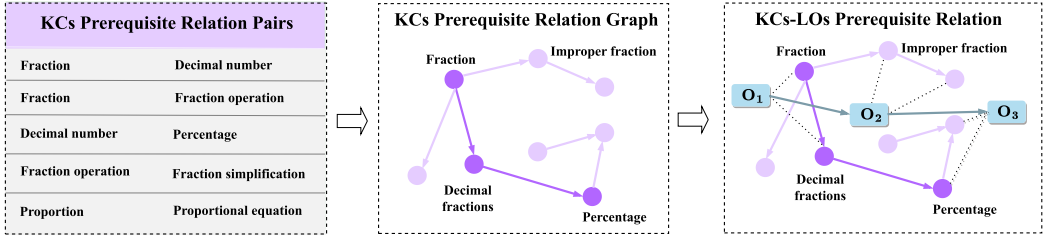


Fig. 4. The transformation process of prerequisite relations: (a) KCs-prerequisite relation pairs show the binary prerequisite pairs, (b) KC-prerequisite relation graph visualizes the prerequisite relations between KCs, (c) KCs-LOs prerequisite relation combines LO with KC, and solid lines (purple, blue) represent prerequisite relations.

while satisfying course-level dependency constraints:

$$\begin{aligned}
 & \min_{A, \varepsilon} \|\text{vec}(A)\|_1 + \lambda \cdot \sum_{i \rightarrow j \in \Omega} \varepsilon_{i \rightarrow j}^2, \\
 & \text{s.t. } x_i^T [C_{i \rightarrow j} \odot A] x_j \geq \theta - \varepsilon_{i \rightarrow j}
 \end{aligned} \tag{4}$$

where both θ and λ are positive constants. ε is a slack variable. This formulation ensures that course dependencies are reflected in KC prerequisite relations while maintaining logical consistency through additional constraints on the prerequisite relations.

Building on this foundational work, Lu et al. [63] proposed two improvements: the **domain-specific concept extraction (DsCE)** method and the **iterative prerequisite relation learning (iPRL)** method. DsCE addressed the limitations of previous methods by first identifying high-quality phrases from educational materials and then evaluating their domain specificity through relationship analysis. The iPRL method enhanced the previous framework [57] by combining optimization-based and learning-based approaches in an iterative process, introducing new content-based features from textbooks and Wikipedia, and developing a unified model that considers both the relationships between KCs and their learning patterns. This combined approach achieved better performance by leveraging both the structural information inherent in educational materials and the semantic features from learning content, while reducing the dependency on manually labeled training data.

While these optimization-based approaches demonstrated the potential of utilizing LO information for KCs' prerequisite relation learning, they were limited by their focus on direct mappings between course dependencies and KC relations, without fully exploiting the rich information contained in learning materials.

3.3.2 Graph-based Models with LO Enhancement. Prerequisite relations between KCs are often sparse and noisy, making it challenging to learn reliable prerequisite patterns from KC pairs alone. A more comprehensive approach is to view these relations as a KC graph rather than isolated KC pairs, which transforms the prerequisite relation task into a link prediction problem on the KC graph. As illustrated in Figure 4, while previous methods might independently predict pairwise relations such as $A \rightarrow B$, $B \rightarrow C$, and $A \rightarrow C$, graph-based approaches can naturally model the sequential chain $A \rightarrow B \rightarrow C$ by considering the global structure. This graph perspective enables **Graph Neural Networks (GNNs)** to capture both local and global dependency patterns [113, 124], which is crucial for understanding the broader knowledge structure and generating coherent learning paths.

In addition, KCs are often embedded in LOs, such as course materials, textbooks, and educational resources. These LOs provide rich contextual information and implicit connections between KCs that can be naturally integrated into the graph structure. By constructing a heterogeneous graph that incorporates both KCs and LOs, researchers can leverage the explicit KC-LO relationships and the implicit KC connections revealed through shared LOs. This heterogeneous graph structure enriches the representation of KCs through their associated learning materials and helps bridge the sparsity gap in direct KC-KC relationships.

To utilize the rich information from LOs in prerequisite learning, Li et al. [52] proposed an unsupervised learning method named the **relational-variational graph autoencoder (R-VGAE)**. By developing a heterogeneous KCs-LOs graph that encompasses both LO-to-LO and LO-to-KC relations, their method effectively leverages the rich contextual information contained in LOs. The VGAE model [45] takes the graph adjacency matrix and node representations as input to identify missing KC-to-KC relations, as shown in Figure 4. This approach demonstrates how the integration of LO information can enhance prerequisite chain learning in a single domain. Building on this foundation, the authors extended their work with the **cross-domain variational graph autoencoder (CD-VGAE)** [54], which enables unsupervised prerequisite chain learning across different domains.

Jia et al. [42] proposed the CPRL model, which first models multiple complex learning resource relations by constructing a KCs-LOs heterogeneous graph, then learns representations of KCs by using R-GCN, and finally predicts prerequisite relations between KCs using a Siamese network. Considering the high cost of obtaining labeled data, the authors proposed two weakly supervised variants: one that learns from learning object dependencies (CPRL_lo), and another that generates probabilistic training data using a data programming paradigm (CPRL_dp). This approach not only makes full use of the complex relationship between KCs and LOs to enhance the KC representation, but also reduces the dependence on large amounts of labeled data through weakly supervised learning. Zhang et al. [121, 122] introduced an end-to-end model named MHAVGAE, which utilizes a KC-LO graph to learn KCs prerequisite relations, incorporating some manually annotated relations. MHAVGAE employs a multi-head attention mechanism to compute the hidden representations of vertices and a gated fusion mechanism to enrich KC features. It then extracts KCs embeddings, inputs its adjacency matrix and features into a VGAE model, and infers prerequisite relations between unannotated KC pairs.

In addition, Mazumder et al. [68] proposed a new framework for prerequisite relation between KCs extraction based on GNNs, called HGAPNet. This framework models heterogeneous graph nodes (KCs, LOs) by representing them through word embeddings and weighting different edges of the heterogeneous graph nodes through a graph attention network. The prediction network combines the KCs representations in the attention network with a nonlinear transformation, thereby predicting prerequisite relations. The main advantages of HGAPNet are its comprehensive modeling of heterogeneous relations, effective attention mechanisms, and end-to-end joint training implementation.

3.3.3 Pre-trained Models with LO Enhancement. While graph-based approaches with LO enhancement have shown good results in capturing structural relationships, they may not fully utilize the rich semantic information contained in LOs. Pre-trained language models like BERT have demonstrated strong capabilities in learning contextual representations. Several approaches attempt to identify KC prerequisite relations by combining the semantic modeling of LOs text content using pre-trained models with the structural relations between KCs and LOs captured through graph structures [51, 93, 97, 104, 126].

A common thread among these approaches is the dual representation strategy. Li et al. [51] developed a dual pre-trained framework that simultaneously utilizes BERT for semantic modeling and Node2Vec for structural representation. BERT encodes latent features from KC descriptions (LOs), while Node2Vec generates discriminative features from KC citation graphs. The two models are jointly optimized to learn representations containing both textual and structural information. However, this approach is limited by its reliance on Wikipedia resources rather than educational materials specifically designed for learning. Building upon this dual representation concept, researchers have expanded the integration of multiple information sources. Wang et al. [104] proposed the HIEPL model, which combines BERT's textual processing with a comprehensive framework that integrates: (1) semantic information from course content; (2) hierarchical co-representation learning across KCs, lecture videos and courses; (3) sequential pattern modeling from video appearances and student viewing behaviors; and (4) heterogeneous graph-based reasoning to capture complex relationships between different LOs.

The challenge of domain-specific knowledge adaptation has prompted innovative solutions. Sun et al. [93] introduced ConLearn, which enhances contextual representations through domain-specific fine-tuning using course corpora. To address the structural aspects, the method constructs KC prerequisite graphs and applies gated graph neural networks. Additionally, recognizing the importance of concept relationships, it employs self-attention mechanisms to fuse information from related concepts that often share similar prerequisite relations. Zhu et al. [126] proposed a novel retrieval-augmented approach to extend pre-trained models capability in handling unseen KCs. Their innovation lies in using KC names to retrieve relevant Wikipedia pages for enriching KC descriptions, and implementing an alternating knowledge distillation strategy where content-based and graph-based models take turns being teacher and student. This iterative process creates a more robust model that can effectively handle previously unseen knowledge components.

These hybrid approaches demonstrate how pre-trained language models, when enhanced with LOs information, can leverage the strengths of both semantic and structural methods, creating more comprehensive solutions for prerequisite relation modeling in educational settings.

3.4 KCs-enhanced Learning for LOs' PR Learning

This section explores how KCs can improve LO prerequisite relation learning. This approach offers two key advantages: First, KCs are more structured knowledge units. These well-defined units serve as bridges to connect different LOs, especially when direct relations between LOs are difficult to establish. Second, since LOs can be decomposed into constituent KCs, their prerequisite relations can be systematically inferred from KC-level dependencies, making the learning process more interpretable and reliable.

Yang et al. [112] proposed an approach that leverages KC dependencies to understand and predict prerequisite relations between courses across different educational contexts. For example, in computer science education, while universities may name and structure their programming courses differently, they often cover similar fundamental KCs like *variables*, *functions*, and *object-oriented programming*. This illustrates how KC-based methods enhance interpretability. When we understand that Course A is a prerequisite for Course B through their shared KCs (e.g., *variables* being necessary before *functions*), the reasoning becomes more transparent than methods that only examine course-level features. Their approach first introduced four methods (Word, SCW, DWE, Cat) to extract and define the KC space from course descriptions. Based on these KCs, they proposed two optimization algorithms: CGL.Class and CGL.Rank, which analyze both the course dependency matrix and the course-KC relation matrix to derive optimal KC prerequisite relations. By leveraging shared KCs, similar prerequisite patterns can be identified between courses at different universities, facilitating course selection across institutions or MOOC platforms. Later work

by Liu et al. [60] further improved the efficiency of the CGLRank algorithm by reducing its time complexity.

Following semantic similarity principles, researchers explored how to sequence LOs based on concept granularity, drawing inspiration from the ‘IS-A’ taxonomy [82]. De Medio et al. [69] developed various metrics, including article length, link density, noun frequency, and hierarchical depth within Wikipedia category structures, to determine the generality-specificity relationship between LOs, thereby establishing prerequisite relations. Subsequent advances came from Manrique et al. [67] and Xiao et al. [110], who investigated direct computation of LO prerequisite relations through their constituent KCs. Both studies quantified LO prerequisite relations by averaging the prerequisite relations among their principal KCs, although they differed in their feature extraction approaches: the former utilized DBpedia data, while the latter employed a transformer-based architecture for KC pair feature learning.

These KC-enhanced approaches demonstrate unique advantages in prerequisite relation learning by providing a structured bridge between different LOs and enabling more transparent reasoning about prerequisite relations. While other methods may excel at capturing direct relationships or specific patterns, KC-enhanced approaches offer complementary benefits through their interpretability and systematic decomposition of learning content. However, their effectiveness depends on the quality and completeness of KC extraction and representation, highlighting the importance of robust knowledge engineering in educational contexts.

4 Datasets and Evaluation

This section presents open datasets for prerequisite relations learning, which can be classified into three types: datasets for KC prerequisite relations, LO prerequisite relations, and comprehensive datasets that cover both. We review key open-access datasets and established methods for dataset construction, with detailed specifications provided in Table 3. The section concludes with a discussion of common evaluation metrics used in prerequisite relation learning.

4.1 KC Prerequisite Relation Datasets

4.1.1 CrowdComp. The CrowdComp dataset [96] was constructed from Wikipedia, comprising five domains: Global Warming, Meiosis, Newton’s Laws, Parallel Postulate, and Public-Key Cryptography. The authors employed **random walk with restart (RWR)** on Wikipedia’s link network to identify 20 key nodes in each domain and extracted up to 400 connected target nodes, resulting in 1,600 KC pairs. These KC pairs were then evaluated through user voting on Amazon Mechanical Turk.³ The limited number of KC pairs in each domain combined with the low proportion of positive samples poses significant challenges for the performance of machine learning models. To address these limitations, subsequent researchers have developed various strategies, including oversampling techniques and the exploitation of prerequisite relation transitivity, to enhance the dataset [34, 40, 55, 67, 88, 125].

4.1.2 RefD Course. RefD course dataset [55] contains KC prerequisite relation pairs extracted from university course descriptions. The authors established course-Wikipedia page mappings through title matching and similarity calculation techniques. The final dataset encompasses 678 KC pairs from computer science and 658 KC pairs from mathematics.

4.1.3 AL-CPL. The AL-CPL dataset [103] was developed for KC prerequisite relation learning from textbooks and subsequently enhanced by Liang et al. [56], who addressed inaccuracies and expanded its coverage. Through the application of transitivity and anti-symmetry principles, the

³<https://www.mturk.com>

Table 3. Prerequisite Relations Learning Datasets

Type	Dataset	Sources	Domains	#KCs/LOs	#KC/LO pairs	#Positive	Language
KC prerequisite relation datasets	CrowdComp [96]	Wikipedia	Global Warming	334	400	43	English
			Meiosis	254	400	67	
			Newton's Laws	292	400	44	
			Parallel Postulate	158	200	25	
	RefD-Course [55]	Course	Public-key Cryp	179	200	27	English
			Computer Science	63	678	108	
	AL-CPL [56, 103] (ITA-PREREQ [70])	TextBook	Math	65	658	75	English (Italian)
			Data Mining	120 (75)	826 (429)	292 (154)	
			Geometry	89 (73)	1,681 (1,338)	524 (430)	
			Physics	153 (131)	1,962 (1,651)	487 (409)	
	MOOC [75]	Course	Precalculus	223 (176)	2,060 (1,504)	699 (502)	English
			Machine Learning	244	7,411	1,735	
University Course [57]	Course	Data Structure and Algorithm	201	5,025	1,148	English	
		Computer Science	569	3,758	1,008		
		Computer Science	345	1,008	1,008		
		Computer Science	487	3,201	3,201		
		Machine Learning	487	3,201	3,201		
LO prerequisite relation datasets	Udacity and edX [35] video-Req [98]	Course Videos	Computer Science, Biology, et al.	6,398	226	113	English English
			Science, math and biology	1,500	2,797	1,113	
	MOOC-PR [110]	Course	Food and health	29	174	13	English
			Algorithms	59	437	96	
			Calculus	63	781	378	
			Global History	51	441	84	
Other comprehensive datasets	TutorialBank [30]	Slide	Natural Language Processing et al.	208	42,643	904	English
	LectureBank 2.0 [52]	Slide	Natural Language Processing et al.	322	68,074	1,551	English
	LectureBankCD [54]	Slide	Computer Vision	201	12,003	871	English
			Bioinformatics	100	8,140	234	
	MOOCCube [115]	Course	Computer Science et al.	700	3,616	1,605	Chinese
	MOOCCubeX [116]	Course	Computer Science et al.	2,134	4,865	2,317	Chinese

enhanced version of AL-CPL contains 6,529 KC pairs across four domains: Data Mining, Geometry, Physics, and Precalculus. Building upon this foundation, Miaschi et al. developed the Italian prerequisite relations dataset ITA-PREREQ to investigate cross-linguistic variations in prerequisite relations [70].

4.1.4 MOOC. The MOOC dataset [75] includes KC prerequisite relations extracted from Coursera course video subtitles, covering courses in Machine Learning, **Data Structures and Algorithm (DSA)**, and **Calculus (CAL)**. The authors extracted subtitles using the Coursera-dl tool,⁴ systematically identified key KCs, and established prerequisite relations between these KC pairs through expert voting. The dataset encompasses 2,883 KC prerequisite pairs derived from 997 videos across 13 machine learning and DSA courses. Note that the CAL data is not publicly available.

4.1.5 University Courses. The University Course dataset [57] contains KC prerequisite relations extracted from computer science department courses across 11 U.S. universities. The dataset construction followed a systematic process: first, the extraction of course-related data including IDs, names, descriptions, and dependencies; second, the implementation of the Wikipedia Miner Toolkit [71] to identify Wikipedia concepts within these descriptions, establishing key KCs for each domain. The process concluded with a voting procedure to confirm relations between KC pairs. The resulting dataset encompasses 654 courses, comprising 639 identified pairs of course dependencies and 1,008 KC pairs with validated prerequisite relations.

4.1.6 NPTEL MOOC. The NPTEL MOOC dataset [85] was developed from MOOCs hosted on the Indian online education platform NPTEL.⁵ This dataset incorporates 382 videos from 38

⁴<https://github.com/coursera-dl/coursera-dl>

⁵<https://npTEL.ac.in>

computer science courses, with subtitle text extraction and subsequent annotation yielding 1,455 pairs of video dependencies and 1,008 pairs of KC prerequisite relations. Notably, the selected KC pairs align with those in the University Course dataset, focusing exclusively on the computer science domain.

4.1.7 Metacademy. The Metacademy dataset [88] originates from the open-source learning platform Metacademy. The construction process began with the extraction of expert-annotated machine learning KCs and their prerequisite relations from the platform. The authors then employed the DuckDuckGo search engine⁶ to establish mappings between these concepts and Wikipedia articles, resulting in 3,201 pairs of KC prerequisite relations.

4.2 LO Prerequisite Relation Datasets

The relationship between KCs and LOs exhibits significant sparsity in educational settings, with distinct KCs being considerably fewer in number than LOs. This disparity poses substantial challenges in accurately labeling prerequisite relations among the numerous LOs. However, several datasets have been established focusing on LO prerequisite relations.

4.2.1 Udacity and edX. The Udacity and edX dataset [35] was extracted from course videos on the online education platforms Udacity and edX. The collection encompasses 5,597 LOs from Udacity and 801 from edX, with manually annotated prerequisite relations totaling 132 and 94 pairs, respectively.

4.2.2 MOOC-PR. MOOC-PR [110] is a course-level LO prerequisite relation dataset, constructed from videos across four Coursera courses: Food and Health, Algorithms, Calculus, and Global History. The dataset features expert-validated LO prerequisite relations established through a voting process, containing 1,833 videos and 571 pairs of LO prerequisite relations.

4.2.3 Vid-Req. The Vid-Req dataset [98] focuses on academic video prerequisite relations, incorporating over 1,500 animated videos from Extramarks⁷ across science, mathematics, and biology curricula for grade 6 through grade 12. Initially containing over 1.1 million video pairs, the dataset was systematically refined using chapter-level prerequisite relations, yielding 2,797 hand-labeled edges, distributed between 1,113 prerequisite relations and 1,684 non-prerequisite relations.

4.3 Other Comprehensive Datasets

The proliferation of online learning platforms has generated vast amounts of learning data, encompassing crucial information about KCs, LOs, courses, and users' learning logs. These datasets incorporate not only structured user and learning data, but also expert-annotated KC prerequisites.

4.3.1 TutorialBank and LectureBank. The TutorialBank dataset [30] supports educational research in NLP, offering 6,300 learning resources classified and annotated across 305 topics. To enhance accessibility, the authors also provided a dedicated search engine and a command-line tool. Domain experts conducted dependency annotations among 200 topics, identifying 794 pairs of topic dependencies. Subsequently, the LectureBank dataset emerged as an enhancement to TutorialBank [53], comprising 1,352 slide resources from 60 courses across five domains: NLP, machine learning, artificial intelligence, deep learning, and information retrieval from various US universities. With its 208 topics and 921 labeled pairs of topic prerequisite relations, LectureBank provides

⁶<https://duckduckgo.com>

⁷<https://www.extramarks.com>

finer knowledge granularity than TutorialBank, enhancing its suitability for prerequisite relation tasks. The dataset primarily aims at leveraging topic prerequisite relations in search engines to optimize search result sequencing by learning order, addressing key educational challenges. Further developments of the LectureBank dataset have continued through subsequent research [52, 54].

4.3.2 MOOCCube and MOOCCubeX. The MOOCCube dataset [115] advances research in the realm of MOOCs. This comprehensive dataset integrates information on courses, concepts, and student engagement. The authors collected 706 courses across 24 disciplines from the Chinese MOOC platform XuetangX, encompassing over 38,000 video segments. Analysis of video subtitles yielded approximately 100,000 course concepts and 17,686 concept prerequisite relations, with 3,616 manually annotated and the remainder identified through automated processes. As an extension, Yu et al. introduced MOOCCubeX, an enhanced version containing 4,216 courses and 230,000 video segments [116]. This expansion facilitated the extraction of 637,000 concepts from video subtitles and the annotation of 1.267 million concept prerequisite relations, with 4,865 obtained through manual annotation and the remainder generated via machine learning models. These annotated relations served to construct course concept maps. Additionally, MOOCCubeX incorporates data from 3.33 million students, 358,000 exercises, and 296 million learning activities, substantially expanding its value for MOOC research.

4.4 Evaluation

Evaluating prerequisite relation learning plays a crucial role in validating identification methods and ensuring the practical applicability of discovered prerequisite relations in educational contexts. This evaluation process encompasses two categories: intrinsic evaluation using standard classification metrics and extrinsic evaluation involving domain experts and educational stakeholders.

Standard classification metrics such as Accuracy, Precision, Recall, F1 Score, and AUC provide quantitative assessment of prerequisite relation prediction performance [55, 96]. These complementary metrics help evaluate different aspects of model effectiveness, from overall correctness to specific prediction capabilities.

Extrinsic evaluation, on the other hand, incorporates expertise and educational perspectives to ensure that the identified prerequisite relations are not only statistically sound but also pedagogically meaningful. One prominent approach involves domain experts reviewing prerequisite relations, with inter-rater agreement measured by Fleiss' Kappa κ [31]. The structured manual annotation process begins with detailed annotation guidelines defining prerequisite relations criteria, followed by a pilot phase for guideline refinement and independent expert evaluation of concept pairs with documented justifications. This methodology has proven effective in label annotation and unsupervised methods evaluation [57, 86, 96], where multiple experts collaboratively establish ground truth through consensus-based resolution of disagreements. Moreover, teacher and student feedback offers crucial insights into the practical relevance of identified prerequisites. Teacher evaluations employ metrics such as content validity ratio and expert consensus scoring. Student-based evaluations focus primarily on learning effectiveness indicators, including prerequisite mastery-performance correlations and comparative pass rates among students with varying levels of prerequisite knowledge. These stakeholder-based assessments provide vital validation of prerequisite relations in real-world learning contexts, complementing traditional statistical metrics.

5 Challenges and Future Opportunities

5.1 Current Challenges

Despite recent advances in prerequisite relation learning, several critical challenges remain. These challenges can be broadly categorized into technical challenges and pedagogical challenges.

5.1.1 Data Imbalance. The challenge of data imbalance has been partially addressed through various approaches. Sampling techniques such as oversampling minority class examples and employing data augmentation strategies have helped balance the training data [121]. Additionally, weak supervision techniques have enabled the generation of more balanced training data by leveraging structural properties of prerequisite relations. However, significant imbalance remains a persistent challenge across large-scale prerequisite relation datasets. This imbalance particularly impacts the performance of deep learning models and becomes more pronounced when dealing with emerging educational content where prerequisite relations are inherently sparse.

5.1.2 Prerequisite Relation Chain. Graph-based models have made significant progress in modeling prerequisite chains, successfully capturing both local and global dependencies [52, 54, 121]. However, crucial challenges remain in ensuring the logical consistency of longer chains, particularly in: (1) maintaining acyclicity in generated chains; (2) capturing long-range dependencies across different educational modules; (3) validating chain quality. These challenges impact the practical application of prerequisite chains in curriculum planning and learning path generation, where chain validity and completeness are essential for effective learning path design.

5.1.3 Dynamic Learning Environment. Current models for learning prerequisite relations in online education typically operate under a critical assumption: that KCs and LOs remain unchanged after the models are trained and deployed. However, the reality of online education platforms is far more dynamic. The continual introduction of new educational content and evolving KCs create a scenario characterized by feature incrementality [126]. Changes in the dimensions of input data and alterations in learner behavior can undermine the existing static model's ability to maintain accurate predictions amidst dynamically changing educational content.

5.1.4 Pedagogical Practice Integration. Current research focuses on computational metrics and statistical evaluations, often neglecting how these relations translate into actual learning outcomes. This issue is compounded by the fact that prerequisite relations are not universal, as their application can vary slightly based on the pedagogical approaches adopted by teachers or educational material authors. For example, some teachers may present broader concepts first, followed by more specific ones, while others may reverse this sequence. This pedagogical flexibility, combined with the inherent complexity of human learning processes, adds another layer of difficulty in generalizing and validating the effectiveness of prerequisite relations across diverse educational settings.

5.2 Future Opportunities

The challenges discussed above point to several promising research directions that could significantly advance the task of prerequisite relation learning.

5.2.1 Combination of Oversampling and Undersampling. In existing work, one common method to address the problem of data imbalance is oversampling. This technique focuses on balancing the ratio of positive to negative samples by increasing the number of positive instances. In the task of identifying prerequisite relations, oversampling can be implemented using various methods, such as leveraging transitive positive samples and synthesizing new positive samples [21] (e.g., employing the SMOTE algorithm). However, oversampling poses several potential issues [119]. For instance, transmitting positive samples may lead to overfitting, and synthesizing samples could introduce noise into the dataset. Exploring a combination of both oversampling and undersampling to minimize the effects of data imbalance represents a potential research direction.

5.2.2 Cycles Detection and Elimination. Future research should focus on developing robust methods for detecting and eliminating cycles in prerequisite chains. Current graph-based

approaches could be enhanced with constraint mechanisms that ensure acyclicity during the learning process. In future work, evaluation metrics such as the **directed acyclic graph (DAG)** rate and **feedback arc set (FAS)** can be used to detect cycles and assess their severity [114]. These metrics will not only identify the presence of cycles but also quantify the modifications needed to ensure the correctness and reliability of prerequisite relations. These technical improvements would directly support educational applications like learning path generation and curriculum planning by ensuring valid knowledge progression sequences.

5.2.3 Enhancing Prerequisite Relation Learning with LLMs. Unlike traditional machine learning and deep learning models, **large language models (LLMs)** such as ChatGPT [18, 80, 81], Llama [100], and ChatGLM [123] might potentially be adapted for more dynamic learning approaches. This potential capability could allow them to update their understanding and predictions based on new data without requiring a complete retraining cycle. By exploring techniques like few-shot or Chain-of-Thought prompting [105, 106], LLMs might be able to adapt to new KCs and LOs as they emerge. This approach could potentially help improve the relevance and accuracy of prerequisite relation models over time. The integration of LLMs into online learning platforms could potentially offer new ways to address the challenge of adapting to new knowledge and learner behaviors, and might contribute to the personalization and responsiveness of educational tools.

5.2.4 Improved Prerequisite Relation Application. Prerequisite relations have shown some potential as a component in educational research and applications. These relations could potentially help in improving the sequencing of learning resources, which might support learners in building foundational knowledge before progressing to more advanced topics [8, 25, 38]. They may play a role in KT, possibly improving the prediction of student learning outcomes by considering the dependencies between KCs [1, 23, 102]. Additionally, prerequisite relations could potentially aid in conducting cognitive diagnoses by helping identify underlying knowledge gaps, which may enable educators to provide more targeted interventions [43, 65]. Through the integration of prerequisite relations into these tasks, research suggests that we could potentially enhance learning experiences, personalize educational content, and support more adaptive teaching strategies.

6 Conclusion

In this survey, we explored the taxonomy of learning resources, along with the definition and attributes of prerequisite relations. To provide a systematic framework for understanding prerequisite relation learning methods, we introduced clear classification criteria based on both feature types and enhancement relationships between KCs and LOs. Following these criteria, we categorize existing methodologies into four distinct types and provide a comprehensive analysis of their merits and limitations. Additionally, we present a detailed overview of available open-access datasets and evaluation metrics for prerequisite relation learning.

Prerequisite relation learning has profound implications for various educational applications. At the micro level, it enables adaptive learning path generation that personalizes content sequencing based on individual learner's knowledge state and learning pace. At the macro level, it assists curriculum designers in organizing course materials and structuring educational programs more effectively. Moreover, prerequisite relations support intelligent tutoring systems in providing targeted learning resource recommendations and help identify potential learning bottlenecks through knowledge gap analysis. These applications collectively contribute to the broader goal of enhancing educational effectiveness and learning outcomes in both traditional and online learning environments.

Furthermore, we discuss key challenges in the prerequisite relation learning task that need to be addressed to enhance personalized learning experiences and facilitate the development

of intelligent educational systems. This survey aims at serving as a reference for researchers in prerequisite relation learning, fostering the development and implementation of advanced prerequisite relation learning technologies in educational settings.

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